

28.2 A 12V-Input 1V-1.8V-Output 94.7%-Peak-Efficiency 685A/cm³-Current-Density Hybrid DC-DC Converter with a Charge Converging Phase

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Large voltage-conversion-ratio (VCR) DC-DC converters (e.g. 12V to 1-to-1.8V) with high efficiency and high current density are essential for industrial and automotive applications. Conventional buck converters have to use high-voltage transistors that incur large power loss and bulky inductors that occupy significant board area, leading to compromised efficiency and current density. Recent advancements in hybrid converters [1–6] provide promising solutions for such applications. These converters introduce flying capacitors to lower the voltage stress of the power transistors or to reduce the inductor current (I_L). For instance, double step-down (DSD) related converters [1, 2] and a switched-capacitor buck converter [3] are proposed to lower the voltage stress of the power transistors to up to $V_{IN}/4$, allowing the use of low-voltage transistors for improved efficiency. However, the inductors still have to handle the entire output current (I_O) as that in a conventional buck converter, requiring bulky inductors with low DCR and thereby limiting current density. Dual-path hybrid converters are proposed in [4–6] to reduce I_L by leveraging the flying capacitors to provide additional current paths. However, the reduction in I_L is relatively modest, only 20% in [4, 5] for a 12V-to-1.2V conversion scenario. Although [6] achieves a significant reduction of 40%, it requires a complex switched-capacitor network with 12 switches. In this paper, we propose a dual-path hybrid buck converter featuring a charge converging phase (CCP-HB) that consistently reduces I_L to less than $0.5I_O$.

Figure 28.2.1 reviews the principle of I_L reduction in dual-path buck converters. The flying capacitor C_F is soft charged by the inductor L during phase 1, accumulating a charge Q_C of $I_L T$ or $I_L(1-D)T$, where D represents the duty cycle of the inductor energizing phase and T is the switching period. In phase 2, Q_C is transferred to the output in parallel with L. The ratios of I_L/I_O can subsequently be derived with one or two flying capacitors. Clearly, to achieve maximum reduction of I_L , Q_C should be maximized and thus a charge converging phase using two flying capacitors C_F and C_{FCC} is proposed in this work. In the proposed configuration, C_F is charged by L during phase 1, while C_{FCC} is charged by both L and C_F during phase 2. By reusing Q_C from C_F , the charge transferred to the output V_O from C_{FCC} (Q_{CC}) is increased to $I_L T$, in contrast to $I_L T$ or $I_L(1-D)T$ in prior works. This process results in up to 33%–50% decrease in the I_L/I_O ratio across the entire D range when compared with other works. A dual-path hybrid converter topology with a VCR range of 6.7 to 12, built upon the principle of the charge converging phase, is also proposed, which consists of 1 inductor, 4 flying capacitors (C_{F1} to C_{F3} and C_{FCC}) and 8 switches (S_{L1} to S_{L4} and S_{H1} to S_{H4}).

Figure 28.2.2 illustrates the operation principle of the proposed converter that encompasses two phases: Φ_1 and Φ_2 . During Φ_1 , S_{H2} , S_{H4} , S_{L2} , and S_{L4} are turned on. L is energized, C_{F2} is charged and C_{F1} , C_{F3} , and C_{FCC} are discharged. Meanwhile, the charge Q_{CC} accumulated on C_{FCC} during Φ_2 gets transferred to V_O . During Φ_2 , S_{H1} , S_{H3} , S_{L1} , and S_{L3} are turned on. L is de-energized, C_{F2} is discharged and C_{F1} and C_{F3} are charged. Importantly, C_{FCC} is charged through L, C_{F1} , and C_{F3} concurrently to accumulate $Q_{CC}=I_L T$, thus defining Φ_2 as the charge converging phase. It can be observed that the voltage across C_{FCC} (V_{FCC}) and C_{F1} (V_{F1}) are V_O and $V_{IN}-2V_O$, respectively. The voltage across C_{F2} (V_{F2}) and C_{F3} (V_{F3}) are then derived as $(V_{IN}+V_O)/2$ and $(V_{IN}-3V_O)/2$, respectively. Therefore, V_{SW4} switches between $-(V_{IN}-5V_O)/2$ during Φ_1 and $2V_O$ during Φ_2 . The VCR is derived as $D/(4+D) \in (0, 0.2)$ using inductor voltage-second balance principle. The I_L/I_O ratio is also derived as $2/(4+D) \in (0.4, 0.5)$. Compared with the prior works using one inductor [3, 5, 6] at $V_{IN}=12V$, the proposed converter reduces I_L to always less than $0.5I_O$ over the entire V_O range. This reduction in I_L significantly alleviates the requirement on the DCR of L. As a result, even 2016 series inductors, measuring $2.0 \times 1.6 \text{ mm}^2$, with a 30% I_{sat} of 3A can be used to deliver 5A I_O by the proposed converter, greatly enhancing current density while maintaining high efficiency. To evaluate the efficiency improvement of the proposed topology, the conduction loss of different topologies [3, 5, 6] is studied. For a fair comparison, the switch sizing strategy suggested in [7] is adopted, and a compact inductor with a typical DCR of 50m Ω is used. When normalized by the conduction loss of a conventional buck converter (P_{con_CB}) using 12V devices with on-resistances of 50m Ω (R_{on_CB}) for both the high-side and the low-side power transistors, the proposed topology achieves a loss reduction of up to 61% compared with the prior works, showcasing the best performance among the existing 12V power converters.

Figure 28.2.3 shows the overall structure of the proposed converter including the voltage-mode controller and the specific power rails assigned to the gate drivers of the switches. All switches are implemented by NMOS transistors to save silicon area. S_{H1} to S_{H4} , and S_{L1} are 5V devices (can withstand a voltage of 5.5V) while S_{L2} to S_{L4} are 1.8V devices.

S_{H1} and S_{H2} are turned on alternately and are powered by V_{IN} . S_{H3} and S_{H4} are powered by V_{SW1} and V_{SW2} , with a gate-voltage swing of $(V_{IN}-V_O)/2$. S_{L2} to S_{L4} are powered through an external supply V_{DDL} of 1.8V, and the corresponding bootstrap circuit is implemented to generate appropriate drive voltages. Note that all bootstrap capacitors C_{BST1} to C_{BST3} are integrated on chip. S_{L1} , with its source switches between $-(V_{IN}-5V_O)/2$ and $2V_O$, is driven by V_{SW3} that switches between V_O and $(V_{IN}+V_O)/2$. As a result, the PWM signal from the controller that oscillates between 0 and V_{DD} ($=5V$) needs to be shifted down during Φ_1 and up during Φ_2 . To meet this requirement, a dual-channel level shifter modified from [8] is proposed. The main channel is responsible for voltage shifting between voltage domains and provides a hold-state current. When transitioning between Φ_1 and Φ_2 , the charge stored in the C_{GS} of diode-connected transistors M_{N1} or M_{N2} in the auxiliary channel can be reused as a trigger current I_{trig} to reduce the propagation delays.

The proposed hybrid converter was fabricated in a 0.18 μm BCD process with a die area of 5mm². The power stage operates at 1MHz with one 1 μH inductor, three 10 μF flying capacitors (C_{F1} to C_{F3}), one 22 μF flying capacitor (C_{FCC}) and one 22 μF output capacitor. Figure 28.2.4 shows the waveforms of each switching node during the start-up process for the scenario of 12V-to-1V conversion, indicating the voltage stress across each switch is within the safe-operation region. For $V_O=1.2V$ and $I_O=2A$, the average I_L of the proposed converter is reduced to 0.9A, showing a reduction of 55%. This result aligns with our previous analysis. The load transient with an I_O step of 2A/200ns is also measured to verify the stability of the converter.

Figure 28.2.5 shows the measured efficiency of the proposed converter under various V_O and I_O conditions. Using a large-volume inductor with a small DCR of 10.5m Ω , the peak efficiency reaches 95.4% with $V_O=1.6V$ and $I_O=1A$. The efficiency of the converter maintains above 82.7% for an I_O range of 0.4A to 5A. Due to the I_L reduction by introducing the proposed charge converging phase, a compact inductor of 2016 series ($2 \times 1.6 \times 1 \text{ mm}^2$) with a large DCR of 57m Ω can be used to enhance the current density, and the converter still achieves high efficiency (94.7% peak efficiency and >82.1% over entire 5A range when V_O is 1.2V or higher). In comparison to the prior works, this converter achieves either the highest efficiency or the highest current density by choosing appropriate inductors. With a bulky inductor, similar to [5], our peak efficiency surpasses that in [5], which is the best one among the prior works. When using a compact inductor, this converter not only achieves the highest current density but also maintains higher efficiency than [3, 4, 6].

Figure 28.2.6 summarizes and compares the performance of the proposed converter with state-of-the-art designs. Owing to the proposed topology with a charge converging phase, which enables the use of a compact 2016-series inductor, the proposed converter achieves a power density of 685A/cm³. This is at least 1.96 $\times$ higher than those in the previous designs. Moreover, the concept of using a charge converging phase can be extended to incorporate more flying capacitors, potentially further enhancing efficiency and current density. Figure 28.2.7 shows the die micrograph of the converter alongside a photo of its PCB solution.

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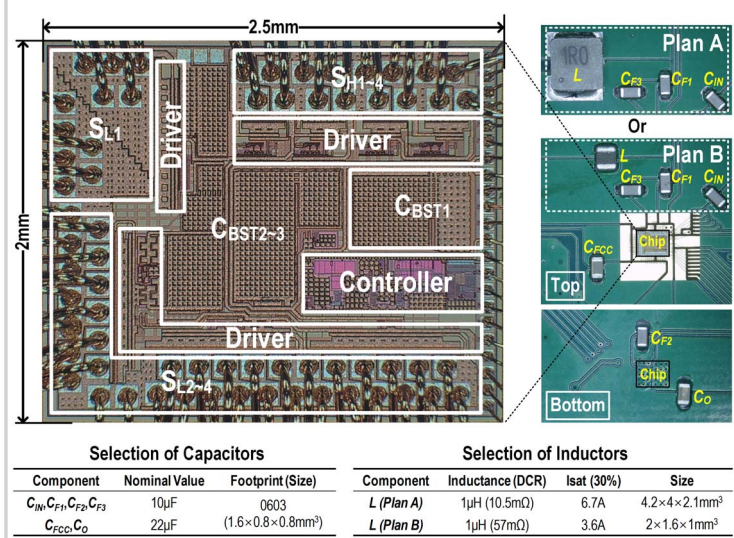


Figure 28.2.7: Die micrograph and a photo of the PCB of the proposed converter.